Transformer-Koopman: History-Conditioned Low-Rank Transition Modulation with Rollout-Aware Training and Real-Time Predictive Control

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***Abstract*—Koopman-based modeling provides a structured framework for predicting nonlinear dynamics through linear evolution in a lifted space. However, a fixed lifted transition may be insufficient for systems whose observed evolution depends on recent trajectory history. This paper proposes a history-conditioned Transformer–Koopman framework that combines temporal representation learning with adaptive lifted dynamics. A patch-based Transformer encoder maps finite observation histories into state-preserving lifted representations, while a context-dependent low-rank correction modulates a nominal transition operator using recent state–input histories. The resulting predictor retains linear latent propagation for a given context while allowing the transition dynamics to vary across histories. To address the mismatch between one-step training and recursive prediction, a two-stage learning procedure is developed, combining nominal initialization with rollout-aware fine-tuning under recursively updated contexts. A finite-horizon analysis characterizes latent rollout inconsistency and the propagation of prediction errors. Finally, the learned predictor is incorporated into a real-time iteration model predictive control scheme that updates the context along predicted trajectories without repeatedly evaluating the Transformer encoder within the prediction horizon.**

***Index Terms*—Koopman operator, Transformer, history-conditioned dynamics, low-rank modulation, rollout-aware training, model predictive control.**

I. Introduction

Learning predictive models for nonlinear controlled systems is a fundamental task in system identification and model-based control [4]–[6]. For predictive control, a useful model must not only capture the system's evolution but also provide a structure that can be efficiently incorporated into an optimization-based controller. Achieving both objectives remains challenging when the available measurements do not fully characterize the underlying dynamics.

Koopman-based modeling offers a structured approach by representing nonlinear dynamics through linear evolution in a lifted space of observables [1], [16]. Finite-dimensional approximations of this representation have been widely explored for prediction and control [17], [29]. However, their effectiveness depends on the choice of observables and the ability of the identified transition model to represent the dynamics encountered during prediction.

Conventional extended dynamic mode decomposition (EDMD) and several learned-lifting approaches typically construct observables from the instantaneous state and identify a fixed lifted transition [8], [18], [20]. When the measured state does not provide sufficient information about recent system evolution, incorporating temporal history can improve the representation. Delay-coordinate methods and history-aware learning architectures address this issue by enriching the information available to the predictor [11], [12]. Nevertheless, incorporating history into the lifting map does not, by itself, remove the restriction of a single global lifted transition. Different histories may produce different lifted states, but their evolution is still governed by the same operator. History-conditioned dynamics provide an alternative by allowing the transition to depend on recent trajectory information [14], [15]. This motivates a model that uses history both to construct the lifted representation and to adapt the transition, without abandoning the linear propagation structure of Koopman-based prediction.

A further difficulty arises in recursive multi-step prediction. A predictor trained using one-step transitions constructed from recorded histories is not necessarily trained on the latent states and histories generated by its own recursive rollout. In a history-conditioned model, this discrepancy affects not only the propagated state but also the context used to determine subsequent transition operators. Consequently, accurate one-step predictions do not automatically imply consistent long-horizon behavior. The training procedure should therefore account for the same recursive propagation and context updates used during deployment.

Motivated by these considerations, this paper develops a history-conditioned Transformer–Koopman framework. A patch-based Transformer encoder maps a finite observation history to a state-preserving lifted representation. A separate context mapping extracts information from recent state–input histories and modulates a nominal lifted transition through a low-rank correction. The resulting predictor allows the transition operator to vary with the recent trajectory while retaining linear propagation in the lifted state and current input for a fixed context. To address the mismatch between one-step training and recursive deployment, a two-stage training procedure combines nominal predictor initialization with rollout-aware fine-tuning under recursively updated contexts.

The learned predictor is further incorporated into a real-time iteration model predictive control (RTI-MPC) framework. Within each prediction horizon, the lifted state is propagated recursively and the transition context is updated from the

predicted histories, avoiding repeated evaluation of the full Transformer lifting encoder. In addition, a finite-horizon analysis characterizes latent rollout inconsistency and prediction-error propagation, while the local approximation involved in the RTI linearization is examined under appropriate regularity conditions.

The main contributions of this paper are summarized as follows:

1) **History-conditioned low-rank Koopman modeling.** We develop a state-preserving temporal lifting model that combines a Transformer encoder with a history-conditioned low-rank modulation of a nominal lifted transition. This construction separates temporal representation learning from transition adaptation while preserving conditional linearity in the lifted dynamics.
2) **Rollout-aware training and finite-horizon consistency analysis.** We formulate a two-stage learning procedure that accounts for recursive latent propagation and context rescheduling during multi-step prediction. A finite-horizon analysis characterizes the discrepancy between recursively propagated and encoder-consistent latent states and establishes conditional bounds on prediction-error propagation.
3) **Real-time predictive control integration.** We incorporate the history-conditioned predictor into a one-iteration RTI-MPC scheme that propagates lifted states and updates transition contexts along the prediction horizon without repeatedly evaluating the full Transformer encoder. The controller formulation includes local linearization of the augmented prediction dynamics and an associated approximation-error characterization.

The remainder of this paper is organized as follows. Section II reviews related work on Koopman-based modeling, temporal representations, and predictive control. Section III introduces the necessary preliminaries and formulates the learning problem. Section IV presents the proposed Transformer–Koopman architecture and its two-stage training procedure. Section V develops the finite-horizon prediction-consistency analysis. Section VI describes the real-time predictive control formulation and its RTI implementation. Section VII presents the experimental evaluation, and Section VIII concludes the paper.

II. Related Work

A. History-Based Koopman Representation

Koopman operator theory describes nonlinear dynamical systems through linear evolution of observable functions [1]–[3]. Finite-dimensional approximations, including DMDc and Koopman-based control models, provide structured predictors for control-oriented applications [26]–[29]. Extended dynamic mode decomposition (EDMD) constructs such approximations using prescribed observable dictionaries [8], [9], [34], while learned-lifting approaches optimize the observable representation from data [18]–[21].

When instantaneous measurements do not sufficiently characterize the system dynamics, temporal information can be incorporated through delay-coordinate or Hankel-based representations [10]–[13]. Recurrent architectures provide an alternative means of encoding temporal dependencies and have also been explored within Koopman-based models: temporal autoencoders combine recurrent layers such as long short-term memory (LSTM) and gated recurrent units (GRU) with Koopman lifting for identification and predictive control [52], and deep recurrent Koopman operators address time-delay systems with robust predictive control [51]. Transformer architectures model temporal interactions through attention, with patch-based tokenization organizing local temporal patterns [35], [36]. These approaches enrich history-dependent observables, but the use of temporal information in the lifting map does not by itself imply a history-conditioned lifted transition.

B. Context-Dependent Koopman Dynamics and Recursive Prediction

Context-dependent Koopman formulations allow the effective transition to vary with operating conditions or recent observations. Koopa employs context-aware operators for non-stationary forecasting [14], while MamKO considers Koopman transitions generated from temporal context [15]. Related Transformer–Koopman architectures have also been investigated for time-series forecasting [37], [38]. Low-rank, state-dependent Koopman transitions have additionally been studied for continuum-robot control [39]. These developments motivate structured alternatives to a single fixed lifted transition. The present work focuses on combining history-based lifting with a context-conditioned low-rank correction to a shared nominal transition, rather than replacing the entire transition with an unrestricted context-dependent matrix.

Recursive prediction introduces a related training challenge. Multi-step objectives have been investigated in learned Koopman models [20], [22]–[24] and neural system identification [25]. Accordingly, multi-step training is not itself the novelty considered here. In a history-conditioned predictor, however, recursive propagation also changes the predicted histories from which subsequent transition contexts are computed. The resulting latent states and contexts may differ from those encountered during one-step training. This motivates training with the same recursive context-update mechanism used for prediction.

C. Real-Time Predictive Control

Real-time iteration (RTI) methods reduce the computational burden of nonlinear model predictive control by performing a single sequential quadratic programming update at each sampling instant [43], [44]. Their convergence and stability properties have been studied under appropriate assumptions [45], [46]. Related quasi-linear parameter-varying formulations represent nonlinear dynamics through linear models indexed by scheduling variables and employ local or iteratively updated predictions for control optimization [47]–[50].

The proposed controller builds on these established ideas but uses a scheduling context computed from finite state–input histories. During multi-step prediction, these histories are recursively updated using the predictor's own outputs,

coupling the scheduling sequence to the predicted trajectory. The resulting formulation combines context rescheduling with latent-state propagation and a one-iteration RTI optimization step, without repeatedly evaluating the full Transformer lifting encoder within the prediction horizon.

III. Preliminaries and Problem Formulation

In this section, we introduce the controlled Koopman representation and formulate the finite-history prediction problem.

A. Koopman Modeling and Temporal Lifting

Consider a discrete-time nonlinear controlled system

$$\mathbf{s}_{k+1} = f(\mathbf{s}_k, \mathbf{u}_k), \tag{1}$$

where $\mathbf{s}_k \in \mathbb{R}^{n_s}$ is the physical state and $\mathbf{u}_k \in \mathbb{R}^m$ is the control input. The available measurement is

$$\mathbf{x}_k = h(\mathbf{s}_k) \in \mathbb{R}^n, \tag{2}$$

where $h : \mathbb{R}^{n_s} \to \mathbb{R}^n$ is the measurement map. We learn the predictor from measured state–input trajectories and use finite observation histories to capture temporal information not resolved by individual measurements.

For a fixed input $\mathbf{u}$, the controlled Koopman operator acts on an observable $\varphi : \mathbb{R}^{n_s} \to \mathbb{R}$ as [1], [3], [7], [16]

$$(\mathcal{K}_{\mathbf{u}}\varphi)(\mathbf{s}) = \varphi(f(\mathbf{s}, \mathbf{u})). \tag{3}$$

The operator is linear with respect to observables:

$$\mathcal{K}_{\mathbf{u}}(\alpha\varphi + \beta\psi) = \alpha\mathcal{K}_{\mathbf{u}}\varphi + \beta\mathcal{K}_{\mathbf{u}}\psi. \tag{4}$$

Finite-dimensional Koopman models approximate this evolution using a selected set of observables. Since only $\mathbf{x} = h(\mathbf{s})$ is measured, we construct the observables from $\mathbf{x}$; each component $g_j(\mathbf{x})$ corresponds to the physical-state observable $g_j \circ h$. An instantaneous lifting map takes the form

$$\mathbf{z}_k = g(\mathbf{x}_k) = [g_1(\mathbf{x}_k), \dots, g_d(\mathbf{x}_k)]^\top \in \mathbb{R}^d, \tag{5}$$

with the input-affine lifted approximation

$$\mathbf{z}_{k+1} \approx \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k. \tag{6}$$

Here $\mathbf{A} \in \mathbb{R}^{d\times d}$ approximates the evolution of the selected observables, and $\mathbf{B} \in \mathbb{R}^{d\times m}$ represents the control-input contribution. Together they define a finite-dimensional approximation of the controlled Koopman evolution in the chosen lifted coordinates.

To incorporate temporal information, we define the state-history window

$$\mathbf{X}_k = [\mathbf{x}_{k-P+1}, \dots, \mathbf{x}_k]^\top \in \mathbb{R}^{P\times n}, \tag{7}$$

and extend the lifting map to

$$\mathbf{z}_k = g_\theta(\mathbf{X}_k) \in \mathbb{R}^d. \tag{8}$$

This representation incorporates recent measurements into the lifted state. The transition model may additionally depend on the recent trajectory context, as developed in Section IV.

B. Learning Problem and Design Objectives

The available training data comprise measured state–input trajectories

$$\mathcal{D} = \{\tau^{(j)}\}_{j=1}^{N_{\mathrm{tr}}}, \qquad \tau^{(j)} = \{(\mathbf{x}_t^{(j)}, \mathbf{u}_t^{(j)})\}_{t=0}^{T_j}. \tag{9}$$

For each admissible index k, the state history $\mathbf{X}_k$ is defined in (7), and the past-input history is

$$\mathbf{U}_k^{\mathrm{in}} = [\mathbf{u}_{k-P+1}, \dots, \mathbf{u}_{k-1}]^\top \in \mathbb{R}^{(P-1)\times m}. \tag{10}$$

The current input $\mathbf{u}_k$ is applied after the observed history window.

Given the measured state history $\mathbf{X}_k$, the past-input history $\mathbf{U}_k^{\mathrm{in}}$, and a future control sequence $\mathbf{u}_{k:k+H-1}$, the objective is to predict the future measurements $\mathbf{x}_{k+1}, \dots, \mathbf{x}_{k+H}$ over a horizon of H steps. The predictor should capture temporal information from the observed history, account for history-dependent variations in the lifted dynamics, and maintain prediction accuracy during recursive multi-step rollout. The proposed model and its training procedure are developed in Section IV.

IV. History-Conditioned Transformer–Koopman Modeling

In this section, we instantiate the predictor of Section III-B using a patch-based temporal lifting, a history-conditioned low-rank transition, and deployment-consistent scheduled training.

A. Temporal Lifting and Low-Rank Transition

Given the history window $\mathbf{X}_k \in \mathbb{R}^{P\times n}$, we partition it into $N_p = P/p$ contiguous non-overlapping temporal patches of length p, where p is assumed to divide P. The ith patch is denoted by

$$\mathbf{X}_k^{(i)} \in \mathbb{R}^{p\times n}, \qquad i = 1, \dots, N_p. \tag{11}$$

Each patch is vectorized as

$$\mathbf{q}_k^{(i)} = \mathrm{vec}(\mathbf{X}_k^{(i)}) \in \mathbb{R}^{pn}, \tag{12}$$

which is then mapped into the Transformer embedding space through a shared linear layer,

$$\boldsymbol{\phi}_k^{(i)} = \mathbf{W}_p\mathbf{q}_k^{(i)} + \mathbf{b}_p \in \mathbb{R}^{d_{\mathrm{model}}}, \tag{13}$$

where $\mathbf{W}_p \in \mathbb{R}^{d_{\mathrm{model}}\times pn}$ and $\mathbf{b}_p \in \mathbb{R}^{d_{\mathrm{model}}}$. This patch-based tokenization reduces the sequence length from P to $N_p = P/p$ while preserving local temporal structure within each patch.

The token sequence is augmented with sinusoidal positional encoding [35] to incorporate temporal order. With $\mathbf{p}^{(i)} \in \mathbb{R}^{d_{\mathrm{model}}}$ the positional encoding of the ith patch, the augmented tokens are

$$\tilde{\boldsymbol{\phi}}_k^{(i)} = \boldsymbol{\phi}_k^{(i)} + \mathbf{p}^{(i)}, \tag{14}$$

and the resulting token sequence is encoded by an L-layer Transformer encoder as

$$\mathbf{H}_k = \mathrm{TransformerEncoder}\Big(\{\tilde{\boldsymbol{\phi}}_k^{(i)}\}_{i=1}^{N_p}\Big) \in \mathbb{R}^{N_p\times d_{\mathrm{model}}}, \tag{15}$$

where L is the number of encoder layers and $\mathbf{H}_k^{(i)} \in \mathbb{R}^{d_{\mathrm{model}}}$ denotes the contextualized representation of the ith patch.

We aggregate the patch features by mean pooling to obtain a fixed-dimensional history descriptor:

$$\mathbf{r}_k = \frac{1}{N_p}\sum_{i=1}^{N_p}\mathbf{H}_k^{(i)} \in \mathbb{R}^{d_{\text{model}}}. \tag{16}$$

After that, the pooled representation is projected to the history-feature space,

$$\mathbf{h}_k = \mathbf{W}_h\mathbf{r}_k + \mathbf{b}_h \in \mathbb{R}^{d-n}, \tag{17}$$

where $\mathbf{W}_h \in \mathbb{R}^{(d-n)\times d_{\text{model}}}$ and $\mathbf{b}_h \in \mathbb{R}^{d-n}$.

The lifted state is defined as

$$\mathbf{z}_k = [\mathbf{x}_k;\, \mathbf{h}_k] \in \mathbb{R}^d, \tag{18}$$

where the first n components of $\mathbf{z}_k$ are the measured state $\mathbf{x}_k = h(\mathbf{s}_k)$ of Section III-A, and the remaining $d-n$ components correspond to the temporal features $\mathbf{h}_k$. This structure allows exact recovery of the *measured* coordinates through the fixed projection

$$\mathbf{x}_k = \mathbf{C}\mathbf{z}_k, \qquad \mathbf{C} = [\mathbf{I}_n\ \mathbf{0}], \tag{19}$$

where $\mathbf{C}$ extracts the measured coordinates from the lifted state.

Although the temporal lifting provides different latent representations for different histories, a fixed transition $\mathbf{A}$ still requires all such history-dependent latent states to evolve under one global linear operator. To relax this additional restriction without discarding the conditional-linear structure, we decompose the transition into a shared nominal component and a structured history-dependent correction.

Definition 1 (History-Conditioned Low-Rank Koopman Family)**.** *Let the history window be $\mathbf{X}_k$ (7), and let $\mathbf{U}_k^{\text{in}} \in \mathbb{R}^{(P-1)\times m}$ denote the input history over the same window (10). Define the lifted state and context vector as*

$$\begin{aligned}\mathbf{z}_k^E &= g_\theta(\mathbf{X}_k) = [\mathbf{x}_k;\, \mathbf{h}_k],\\ \mathbf{c}_k &= C_\phi(\mathbf{X}_k, \mathbf{U}_k^{\text{in}}) \in \mathbb{R}^{d_c},\end{aligned} \tag{20}$$

where g_θ encodes the state history and $C_\phi : \mathbb{R}^{P\times n} \times \mathbb{R}^{(P-1)\times m} \to \mathbb{R}^{d_c}$ maps the state and input histories to the context vector. The superscript E denotes the encoder-generated latent state, while P denotes the model-propagated latent state. The one-step prediction is

$$\hat{\mathbf{z}}_{k+1}^P = \mathbf{A}(\mathbf{c}_k)\,\mathbf{z}_k^E + \mathbf{B}\,\mathbf{u}_k, \tag{21}$$

with the history-conditioned *transition*

$$\begin{aligned}\mathbf{A}(\mathbf{c}) &= \mathbf{A}_0 + \mathbf{U}\,\mathrm{diag}\big(\gamma_\psi(\mathbf{c})\big)\mathbf{V}^\top,\\ \mathrm{rank}\big[\mathbf{A}(\mathbf{c}) - \mathbf{A}_0\big] &\le r.\end{aligned} \tag{22}$$

Here $\mathbf{A}_0 \in \mathbb{R}^{d\times d}$ is the nominal transition, $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d\times r}$ are low-rank factors, and $\gamma_\psi : \mathbb{R}^{d_c} \to \mathbb{R}^r$ provides the context-dependent coefficients. The input matrix $\mathbf{B} \in \mathbb{R}^{d\times m}$ is shared across contexts, and r is the maximum correction rank.

In recursive prediction, the encoder initializes the latent state once, while subsequent steps propagate $\mathbf{z}^P$ with the context updated from predicted histories, as specified in (24). The resulting lifting pipeline is illustrated in Fig. 1.

B. Training Procedure

The predictor consists of the lifting encoder g_θ, the context map C_ϕ, and the dynamics parameters $(\mathbf{A}_0, \mathbf{U}, \mathbf{V}, \psi, \mathbf{B})$, with a fixed projection $\mathbf{C}$. Training proceeds in two stages: nominal one-step pretraining followed by scheduled-rollout fine-tuning, which accounts for the shift from measured-history conditioning during initialization to predicted-history conditioning during recursive deployment.

For a rollout horizon H_R, each training sample comprises the measured histories $(\mathbf{X}_k, \mathbf{U}_k^{\text{in}})$, the future input sequence $(\mathbf{u}_k, \ldots, \mathbf{u}_{k+H_R-1})$, and the corresponding measured-state targets $(\mathbf{x}_{k+1}, \ldots, \mathbf{x}_{k+H_R})$, all drawn from the same trajectory.

Stage 0 (nominal one-step pretraining). We initialize the encoder g_θ and a time-invariant base operator $(\mathbf{A}_0, \mathbf{B})$ by minimizing the one-step physical prediction error over the encoder-generated latent distribution,

$$\mathcal{L}_{\text{init}} = \sum_k \|\mathbf{C}(\mathbf{A}_0\mathbf{z}_k + \mathbf{B}\mathbf{u}_k) - \mathbf{x}_{k+1}\|_2^2. \tag{23}$$

where

$$\mathbf{z}_k = g_\theta(\mathbf{X}_k).$$

This loss jointly initializes g_θ, $\mathbf{A}_0$, and $\mathbf{B}$ using one-step measured-state targets. Since $\mathbf{C} = [\mathbf{I}_n\ \mathbf{0}]$, it directly supervises only the first n rows of $\mathbf{A}_0$ and $\mathbf{B}$; the remaining rows are learned during Stage 1.

Definition 2 (Rollout-Aware Scheduled Objective)**.** *Starting from the encoder-initialized latent $\mathbf{z}_k^P = g_\theta(\mathbf{X}_k)$, the scheduled rollout is generated for $i = 0, \ldots, H_R - 1$ by*

$$\begin{cases}\mathbf{c}_{k+i} = C_\phi(\hat{\mathbf{X}}_{k+i}, \mathbf{U}_{k+i}^{\text{in}}),\\ \mathbf{z}_{k+i+1}^P = \mathbf{A}(\mathbf{c}_{k+i})\,\mathbf{z}_{k+i}^P + \mathbf{B}\,\mathbf{u}_{k+i},\\ \hat{\mathbf{x}}_{k+i+1} = \mathbf{C}\,\mathbf{z}_{k+i+1}^P.\end{cases} \tag{24}$$

where the windows are then shifted by appending the decoded state and the applied input and dropping the oldest entry, $\hat{\mathbf{X}}_{k+i+1} = \mathrm{shift}(\hat{\mathbf{X}}_{k+i}, \hat{\mathbf{x}}_{k+i+1})$ and $\mathbf{U}_{k+i+1}^{\text{in}} = \mathrm{shift}(\mathbf{U}_{k+i}^{\text{in}}, \mathbf{u}_{k+i})$. The rollout starts from $\hat{\mathbf{X}}_k = \mathbf{X}_k$ and the observed input history $\mathbf{U}_k^{\text{in}}$. The lifting encoder is thus evaluated only at the start of the rollout, while the context is re-scheduled at every step from the history that the predictor itself has produced. With uniform weights $w_i = 1/H_R$, the scheduled objective is

$$\mathcal{L}_R = \sum_k \sum_{i=1}^{H_R} w_i \left\|\mathbf{C}\mathbf{z}_{k+i}^P - \mathbf{x}_{k+i}\right\|_2^2. \tag{25}$$

Stage 1 (scheduled-rollout fine-tuning). We activate the context-conditioned transition (22) and jointly optimize $(\theta, \phi, \mathbf{A}_0, \mathbf{U}, \mathbf{V}, \psi, \mathbf{B})$ by minimizing the scheduled objective (25). The encoder initializes each rollout once; subsequent steps propagate the latent state and update the context using predicted histories according to (24). The projection $\mathbf{C}$ remains fixed.

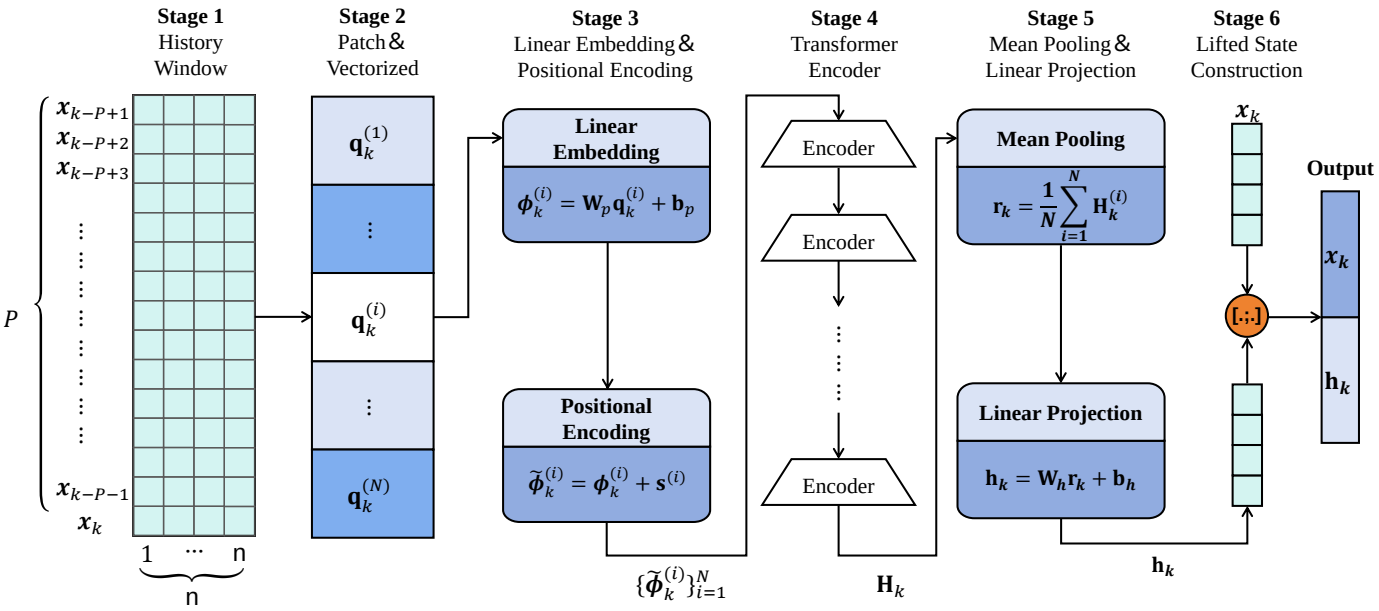


Fig. 1. Encoder pipeline of the history-conditioned Transformer–Koopman predictor. The history window $\mathbf{X}_k$ is partitioned into temporal patches and vectorized; each patch is linearly embedded, augmented with positional encoding, and processed by a Transformer encoder whose token outputs are mean-pooled and linearly projected to a history feature $\mathbf{h}_k$. Concatenating $\mathbf{h}_k$ with the current measurement $\mathbf{x}_k$ forms the state-preserving lifted state $\mathbf{z}_k = [\mathbf{x}_k;\, \mathbf{h}_k]$.

To bound the context-dependent correction, let the columns of $\mathbf{U}$ and $\mathbf{V}$ satisfy $\|\mathbf{U}_{:,i}\|_2, \|\mathbf{V}_{:,i}\|_2 \le 1$, and parameterize the modulation coefficients as

$$\begin{cases} \gamma_{\psi,i}(\mathbf{c}) = g_{\max}\pi_i \tanh\big([\mathbf{W}_g\mathbf{c} + \mathbf{b}_g]_i\big), \\ \pi_i \ge 0, \quad \displaystyle\sum_{i=1}^{r} \pi_i = 1. \end{cases} \tag{26}$$

Here $\boldsymbol{\pi}$ is learned through a softmax parameterization, and the low-rank factor columns are normalized to satisfy the stated bounds. Consequently, $\sum_i |\gamma_{\psi,i}(\mathbf{c})| \le g_{\max}$ and

$$\|\mathbf{A}(\mathbf{c}) - \mathbf{A}_0\|_2 \le \sum_{i=1}^{r} \big|\gamma_{\psi,i}(\mathbf{c})\big| \le g_{\max}, \tag{27}$$

for every context $\mathbf{c}$. This bounds the correction magnitude, not the stability of $\mathbf{A}(\mathbf{c})$.The complete procedure is summarized in Algorithm 1.

Algorithm 1 Two-Stage Transformer–Koopman Training

Input: Trajectory dataset $\mathcal{D}$; history length P; rollout horizon H_R; training epochs E_0, E_1.
Stage 0: Initialize $(\theta, \mathbf{A}_0, \mathbf{B})$ by minimizing (23) for E_0 epochs.
Stage 1: Initialize $(\phi, \mathbf{U}, \mathbf{V}, \psi)$ and activate (22).
for $\ell = 1$ **to** E_1 **do**
  for each mini-batch of length-H_R segments **do**
    Set $\mathbf{z}_k^P = g_\theta(\mathbf{X}_k)$, $\hat{\mathbf{X}}_k = \mathbf{X}_k$, and initialize $\mathbf{U}_k^{\text{in}}$ from the recorded history.
    for $i = 0$ **to** $H_R - 1$ **do**
      Evaluate $\mathbf{c}_{k+i} = C_\phi(\hat{\mathbf{X}}_{k+i}, \mathbf{U}_{k+i}^{\text{in}})$.
      Propagate $\mathbf{z}_{k+i+1}^P = \mathbf{A}(\mathbf{c}_{k+i})\mathbf{z}_{k+i}^P + \mathbf{B}\mathbf{u}_{k+i}$.
      Decode $\hat{\mathbf{x}}_{k+i+1} = \mathbf{C}\mathbf{z}_{k+i+1}^P$ and shift the state and input histories.
    end for
    Update $(\theta, \phi, \mathbf{A}_0, \mathbf{U}, \mathbf{V}, \psi, \mathbf{B})$ using (25).
  end for
end for
Return: Trained encoder, context network, and dynamics parameters.

V. Finite-Horizon Consistency Analysis

In this section, we analyze the internal finite-history predictor of Section IV and its finite-horizon error propagation. Section V-A bounds the inconsistency that arises when the latent state is propagated recursively instead of being re-encoded, and Section V-B bounds the finite-horizon prediction error of the learned augmented predictor against a recorded reference sequence. All statements are conditional on the regularity assumptions stated below and concern the predictor itself; the scope remark at the end of the section states what is *not* claimed.

Because the lifted state is a function of a finite window, the predictor can be recast as a first-order system on an augmented state. Define

$$\boldsymbol{\xi}_k = \mathrm{col}\big(\mathrm{vec}\,\mathbf{X}_k,\ \mathrm{vec}\,\mathbf{U}_k^{\text{in}},\ \mathbf{z}_k\big) \in \mathbb{R}^{Pn+(P-1)m+d}, \tag{28}$$

and consider the predictor's own rollout, in which the newest fed-back state is its decoded output $\hat{\mathbf{x}}_{k+1} = \mathbf{C}\mathbf{z}_{k+1}$. The window update is then a deterministic shift: append the decoded state and the applied input, and drop the oldest entry. Since $\mathbf{X}_k$ and $\mathbf{U}_k^{\text{in}}$ are sub-blocks of $\boldsymbol{\xi}_k$, every argument of that update is a function of $(\boldsymbol{\xi}_k, \mathbf{u}_k)$ alone, so composing the shift, the context map, the transition, and the decoder yields a single deterministic map

$$\boldsymbol{\xi}_{k+1} = \mathbf{F}_\Theta(\boldsymbol{\xi}_k, \mathbf{u}_k). \tag{29}$$

This construction describes the predictor's internal rollout only.

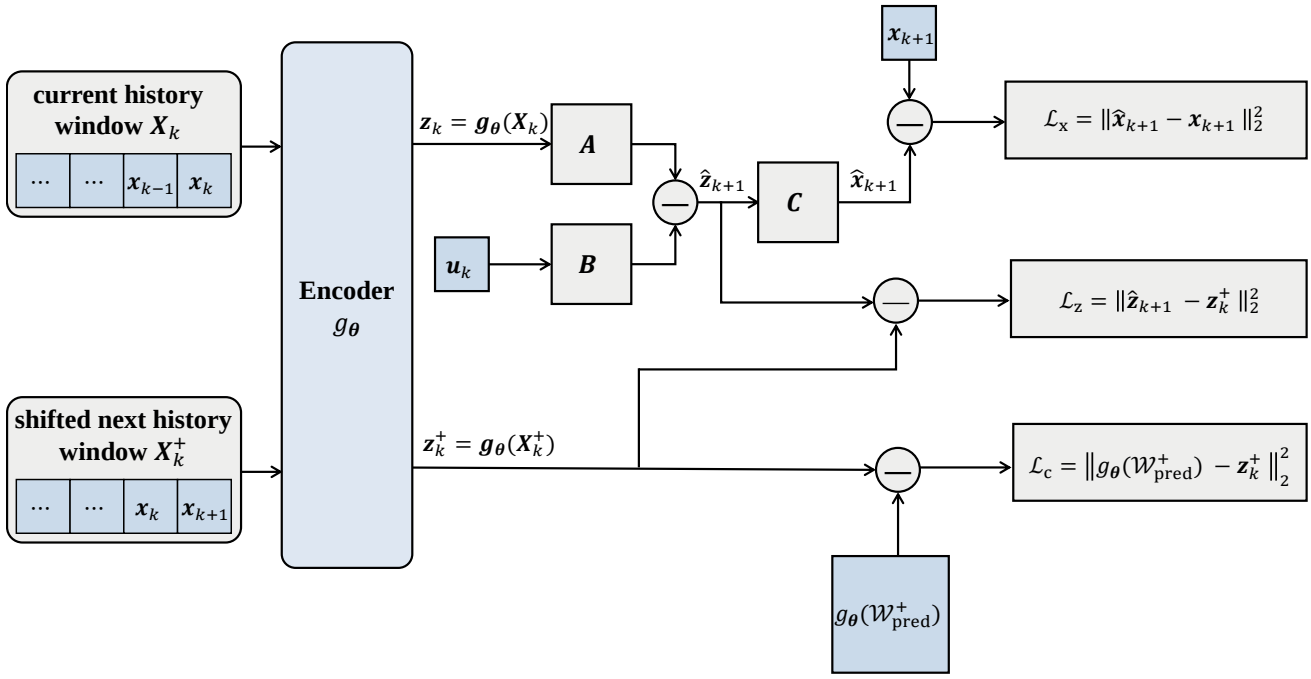


Fig. 2. One-step lifting and prediction (nominal pretraining, Stage 0). The history window $\mathbf{X}_k$ is encoded by g_θ to produce the lifted state $\mathbf{z}_k^E$, and the time-invariant base operator $(\mathbf{A}_0, \mathbf{B})$ produces the one-step prediction $\hat{\mathbf{z}}_{k+1}^P = \mathbf{A}_0\mathbf{z}_k^E + \mathbf{B}\mathbf{u}_k$, from which $\hat{\mathbf{x}}_{k+1}$ is recovered via $\mathbf{C}$. Stage-0 training is supervised only through the decoded physical state, by the one-step loss (23); the shifted window $\mathbf{X}_k^+$ indicates the sliding-window convention over the dataset and does not enter the Stage-0 objective. Stage 1 replaces $\mathbf{A}_0$ by the history-conditioned $\mathbf{A}(\mathbf{c})$ and jointly refines the full model on the rollout-aware scheduled objective (Algorithm 1).

If a true measurement $\mathbf{x}_{k+1}$ is fed back instead, the update also depends on an exogenous quantity and is not a function of $\boldsymbol{\xi}_k$ alone; and nothing here asserts that the finite measured history Markovizes the physical system (1).

A. Latent Rollout Inconsistency and Its Propagation

We now characterize the discrepancy that arises when the latent state is propagated recursively instead of being re-encoded at every step.

Definition 3 (Latent Rollout Inconsistency)**.** *Let $\hat{\mathbf{X}}_{k+i}$ be the history produced by the predictor's own rollout and let $\mathbf{c}_{k+i} = C_\phi(\hat{\mathbf{X}}_{k+i}, \mathbf{U}_{k+i}^{\text{in}})$ denote the scheduled context recomputed from it. Define the encoder-consistent latent and the propagated latent,*

$$\mathbf{z}_{k+i}^E = g_\theta(\hat{\mathbf{X}}_{k+i}), \qquad \mathbf{z}_{k+i+1}^P = \mathbf{A}(\mathbf{c}_{k+i})\mathbf{z}_{k+i}^P + \mathbf{B}\mathbf{u}_{k+i}, \tag{30}$$

and the normalized rollout defect

$$d_{\mathcal{M},k+i} = \frac{\left\|\mathbf{z}_{k+i}^P - \mathbf{z}_{k+i}^E\right\|_2}{\left\|\mathbf{z}_{k+i}^E\right\|_2 + \epsilon_0}, \tag{31}$$

with $\epsilon_0 = 10^{-6}$ a fixed numerical floor. We also use the absolute defect $d_{k+i}^{\text{abs}} = \|\mathbf{z}_{k+i}^P - \mathbf{z}_{k+i}^E\|_2$, which is the quantity for which Proposition 1 provides an upper bound.

Two properties of this defect are worth stating explicitly. First, because $\mathbf{z} = [\mathbf{x}; \mathbf{h}]$ is state-preserving and the newest entry of $\hat{\mathbf{X}}_{k+i}$ is the decoded state $\mathbf{C}\mathbf{z}_{k+i}^P$, the propagated and the encoder-consistent latents share their first n coordinates, $\mathbf{C}(\mathbf{z}_{k+i}^P - \mathbf{z}_{k+i}^E) = \mathbf{0}$; both latents are evaluated on the same predicted history, and the equality does not relate the propagated latent to the encoding of the true history. The defect therefore measures the inconsistency of the remaining $d - n$ hidden lifted coordinates rather than a physical multi-step prediction error. Those hidden coordinates still affect future decoded predictions through $\mathbf{A}(\mathbf{c})$, which is why this inconsistency matters. Second, evaluating $\mathbf{z}_{k+i}^E$ requires an additional encoder pass on the predicted history. That pass is an offline diagnostic used to characterize the mechanism; it is not part of the online rollout, in which the context is re-scheduled without re-running the encoder inside the horizon (Section VI).

Proposition 1 (Latent Rollout Error Propagation)**.** *Let $\mathbf{z}_k^P$ denote the recursively propagated latent state and $\mathbf{z}_k^E = g_\theta(\hat{\mathbf{X}}_k)$ the encoder-consistent latent state, and let $d_k \equiv d_k^{\text{abs}} = \|\mathbf{z}_k^P - \mathbf{z}_k^E\|_2$. Define the encoder-consistent one-step defect*

$$\epsilon_k = \left\|\mathbf{A}(\mathbf{c}_k)\mathbf{z}_k^E + \mathbf{B}\mathbf{u}_k - \mathbf{z}_{k+1}^E\right\|_2. \tag{32}$$

Then $d_{k+1} \le \|\mathbf{A}(\mathbf{c}_k)\|_2\, d_k + \epsilon_k$, and for any horizon H,

$$d_{k+H} \le \left\|\mathbf{A}_{k+H-1}\cdots\mathbf{A}_k\right\|_2 d_k + \sum_{i=0}^{H-1} \left\|\mathbf{A}_{k+H-1}\cdots\mathbf{A}_{k+i+1}\right\|_2 \epsilon_{k+i}, \tag{33}$$

where $\mathbf{A}_{k+j} = \mathbf{A}(\mathbf{c}_{k+j})$ is the transition at the scheduled context $\mathbf{c}_{k+j} = C_\phi(\hat{\mathbf{X}}_{k+j}, \mathbf{U}_{k+j}^{\text{in}})$ along the same rollout. If the rollout starts from an encoder-consistent latent ($\mathbf{z}_k^P = \mathbf{z}_k^E$, hence $d_k = 0$), the first term vanishes and the bound reduces to the second sum alone. The bound is a trajectory-wise recursion: both ϵ_k and the transition matrices $\mathbf{A}_{k+j}$ are

evaluated along the same rollout, so they depend on the realized trajectory rather than being a priori constants.

Proof. Let $\mathbf{e}_k = \mathbf{z}_k^P - \mathbf{z}_k^E$ denote the latent error. Subtracting the encoder-consistent update from the propagation rule $\mathbf{z}_{k+1}^P = \mathbf{A}(\mathbf{c}_k)\mathbf{z}_k^P + \mathbf{B}\mathbf{u}_k$ gives the linear recursion

$$\mathbf{e}_{k+1} = \mathbf{A}(\mathbf{c}_k)\mathbf{e}_k + \mathbf{r}_k^z, \qquad \mathbf{r}_k^z = \mathbf{A}(\mathbf{c}_k)\mathbf{z}_k^E + \mathbf{B}\mathbf{u}_k - \mathbf{z}_{k+1}^E,$$

where $\|\mathbf{r}_k^z\|_2 = \epsilon_k$. Taking norms and using the triangle inequality and submultiplicativity yields the one-step bound $d_{k+1} \le \|\mathbf{A}(\mathbf{c}_k)\|_2\, d_k + \epsilon_k$. Unrolling the recursion over H steps gives

$$\mathbf{e}_{k+H} = \mathbf{A}_{k+H-1}\cdots\mathbf{A}_k\,\mathbf{e}_k + \sum_{i=0}^{H-1} \mathbf{A}_{k+H-1}\cdots\mathbf{A}_{k+i+1}\,\mathbf{r}_{k+i}^z,$$

and taking norms, with $\|\mathbf{e}_k\|_2 = d_k$ and $\|\mathbf{r}_{k+i}^z\|_2 = \epsilon_{k+i}$, gives (33) (the empty product at $i = H-1$ is the identity). Since $\|\mathbf{A}_{k+H-1}\cdots\mathbf{A}_k\|_2 \le \prod_{j=0}^{H-1}\|\mathbf{A}_{k+j}\|_2$, this matrix-product form is at least as tight as the product-of-norms form. □

Equation (33) shows that the long-horizon inconsistency is governed by two factors: the local one-step defect ϵ_k and the *amplification of the time-varying transition products*.

Remark 1 (Why spectral stabilization is insufficient)**.** *Enforcing $\rho(\mathbf{A}(\mathbf{c}_k)) < 1$ for all k does* not *imply a uniform contraction of the time-varying products:*

$$\rho(\mathbf{A}_k) < 1 \;\; \forall k \quad \not\Rightarrow \quad \|\mathbf{A}_{k+H-1}\cdots\mathbf{A}_k\|_2 \le c < 1, \tag{34}$$

which is the classical spectral-radius/joint-spectral-radius gap for non-commuting matrices [40]–[42]. A bound on $\rho(\mathbf{A}(\mathbf{c}))$ therefore does not by itself control the amplification of the scheduled products that appears in (33)*.*

B. Finite-Horizon Prediction Error Bound

To state the bound we compare the learned predictor with a *reference augmented sequence* built directly from the recorded observations, rather than postulating a deterministic reference transition for the plant. Let

$$\boldsymbol{\xi}_k^\star = \mathrm{col}\big(\mathrm{vec}\,\mathbf{X}_k,\ \mathrm{vec}\,\mathbf{U}_k^{\mathrm{in}},\ g_\theta(\mathbf{X}_k)\big) \tag{35}$$

denote the reference augmented state built from the true observations and the encoder-consistent latent, and let $\boldsymbol{\xi}_{k+1}^\star$ be its successor along the recorded trajectory. The reference sequence consists of measured quantities, and no deterministic reference map is assumed to generate it. The learned augmented transition $\mathbf{F}_\Theta$ is the map (29), which propagates the latent and feeds back the decoded output $\hat{\mathbf{x}} = \mathbf{C}\mathbf{z}^P$. We therefore measure the learned model by its one-step discrepancy on the reference sequence,

$$\delta_k^\star = \big\|\boldsymbol{\xi}_{k+1}^\star - \mathbf{F}_\Theta(\boldsymbol{\xi}_k^\star, \mathbf{u}_k)\big\|_2. \tag{36}$$

This quantity bundles the physical modeling error, the finite-history truncation error, the learned transition error, and the difference between the encoder-consistent and the propagated latent updates.

On the recorded sequence we also define the one-step modeling residual at the encoder-consistent latent,

$$\mathbf{w}_k = \mathbf{z}_{k+1}^E - \big(\mathbf{A}(\mathbf{c}_k)\mathbf{z}_k^E + \mathbf{B}\mathbf{u}_k\big), \qquad \mathbf{z}_k^E = g_\theta(\mathbf{X}_k), \tag{37}$$

which measures one-step data consistency on recorded histories. It is not an independent error quantity: because the history and input shifts are exact and the encoder is state-preserving, $\mathbf{C}\mathbf{z}_{k+1}^E = \mathbf{x}_{k+1}$, and the augmented one-step reference discrepancy (36) reduces to

$$(\delta_k^\star)^2 = \|\mathbf{w}_k\|_2^2 + \|\mathbf{C}\mathbf{w}_k\|_2^2, \qquad \|\mathbf{w}_k\|_2 \le \delta_k^\star \le \sqrt{2}\,\|\mathbf{w}_k\|_2, \tag{38}$$

so $\delta_k^\star$ and $\mathbf{w}_k$ are two views of the same one-step discrepancy on recorded data. The identity is exact for the augmented ordering of (35): the history-shift block contributes $\|\mathbf{C}\mathbf{w}_k\|_2$, the latent block contributes $\|\mathbf{w}_k\|_2$, and the input-history block cancels because the reference and the model apply the same input. It uses the history shift, the fixed projection $\mathbf{C} = [\mathbf{I}_n\ \mathbf{0}]$, and the re-encoding of the next recorded state adopted here; it is not an unconditional relation for arbitrary feedback structures. The defect ϵ_k of Proposition 1, by contrast, is evaluated on histories generated by the predictor itself and cannot be replaced by statistics of $\mathbf{w}_k$.

Assumption 1 (Operating-set reference discrepancy)**.** *There exist a compact* operating set $\Omega \subset \mathbb{R}^{Pn+(P-1)m+d} \times \mathbb{R}^m$*, i.e. a set of augmented state–input pairs $(\boldsymbol{\xi}, \mathbf{u})$, and a finite constant $\epsilon \ge 0$ such that, along the reference sequence and at every step of the considered horizon, the one-step reference discrepancy* (36) *satisfies $\delta_k^\star \le \epsilon$, equivalently*

$$\big\|\boldsymbol{\xi}_{k+1}^\star - \mathbf{F}_\Theta(\boldsymbol{\xi}_k^\star, \mathbf{u}_k)\big\|_2 \le \epsilon. \tag{39}$$

We additionally require that the reference and predicted trajectories remain in Ω over the considered horizon; this condition is used explicitly in Theorem 1.

Assumption 2 (Lipschitz continuity of the learned transition)**.** *There exists a finite constant L_{F} such that*

$$\|\mathbf{F}_\Theta(\boldsymbol{\xi}_1, \mathbf{u}) - \mathbf{F}_\Theta(\boldsymbol{\xi}_2, \mathbf{u})\|_2 \le L_{\mathrm{F}}\|\boldsymbol{\xi}_1 - \boldsymbol{\xi}_2\|_2 \tag{40}$$

for all $(\boldsymbol{\xi}_1, \mathbf{u}), (\boldsymbol{\xi}_2, \mathbf{u}) \in \Omega$. No contraction is assumed: L_{F} need not be smaller than one. The bound holds on Ω whenever the context map C_ϕ, the modulation γ_ψ, and the map $\mathbf{c} \mapsto \mathbf{A}(\mathbf{c})$ are Lipschitz there and the lifted state stays bounded; we take these as standing regularity conditions on the learned components. These conditions concern the functions that the recursion (29) *actually composes—the context map, the modulation, the low-rank transition, the affine latent update and decoding, and the exact window shift—and not the lifting encoder: g_θ is evaluated only to initialize the rollout, and is not re-evaluated inside $\mathbf{F}_\Theta$. In the parameterization used here the context map and the modulation are composed of affine layers and smooth activations, so on a bounded operating set these conditions are met.*

Theorem 1 (Finite-Horizon Prediction Error Bound)**.** *For a common prescribed input sequence $\{\mathbf{u}_k, \ldots, \mathbf{u}_{k+H-1}\}$, suppose that the reference and predicted trajectories remain in*

the operating set Ω over the horizon H. Under Assumptions 1 and 2, and with $e_k = \boldsymbol{\xi}_k^\star - \hat{\boldsymbol{\xi}}_k$,

$$\|e_{k+H}\|_2 \le L_{\mathrm{F}}^H \|e_k\|_2 + \sum_{i=0}^{H-1} L_{\mathrm{F}}^{H-1-i}\, \delta_{k+i}^\star. \tag{41}$$

If in addition the reference discrepancy is uniformly bounded by ϵ as in Assumption 1, then

$$\|e_{k+H}\|_2 \le L_{\mathrm{F}}^H \|e_k\|_2 + \epsilon \sum_{j=0}^{H-1} L_{\mathrm{F}}^{j}. \tag{42}$$

If the horizon is initialized from the same measured history and the corresponding encoder-consistent latent state, i.e. $\hat{\mathbf{X}}_k = \mathbf{X}_k$ and $\mathbf{z}_k^P = g_\theta(\mathbf{X}_k)$, then $e_k = 0$ and $\|e_{k+H}\|_2 \le \epsilon \sum_{j=0}^{H-1} L_{\mathrm{F}}^j$. The bound compares the learned predictor with the recorded reference trajectory under a common *input sequence; it is not a closed-loop tracking-error or robustness statement. Here ϵ is the uniform bound of Assumption 1.*

Proof. For one step, using the recorded successor $\boldsymbol{\xi}_{k+1}^\star$,

$$\begin{aligned}
\|e_{k+1}\|_2 &= \big\|\boldsymbol{\xi}_{k+1}^\star - \mathbf{F}_\Theta(\hat{\boldsymbol{\xi}}_k, \mathbf{u}_k)\big\|_2 \\
&\le \big\|\boldsymbol{\xi}_{k+1}^\star - \mathbf{F}_\Theta(\boldsymbol{\xi}_k^\star, \mathbf{u}_k)\big\|_2 \\
&\quad + \big\|\mathbf{F}_\Theta(\boldsymbol{\xi}_k^\star, \mathbf{u}_k) - \mathbf{F}_\Theta(\hat{\boldsymbol{\xi}}_k, \mathbf{u}_k)\big\|_2 \\
&\le \delta_k^\star + L_{\mathrm{F}} \|e_k\|_2,
\end{aligned}$$

by (36) and Assumption 2, which applies at every step because the trajectories remain in Ω over the horizon. Unrolling the recursion yields (41); bounding each $\delta_{k+i}^\star$ by ϵ yields (42). □

Remark 2 (How to read these bounds)**.** *Equation* (41) *is a conditional finite-horizon relation between two trajectories driven by the same input sequence, not a numerical certificate. The quantities ϵ, ϵ_k and $\mathbf{w}_k$ are defined on different trajectories and are not interchangeable, and the regularity constants of Assumptions 1 and 2 are properties of the operating set Ω rather than of any validation sample. The bound concerns the rolled-out predictor under a common input sequence: when $L_{\mathrm{F}} > 1$ it grows geometrically with H, and it is neither a closed-loop tracking-error statement nor a stability result.*

The bound above concerns the augmented state, and the physical coordinates are a sub-vector of it. Both the current physical state $\mathbf{x}_{k+H}$ and the predictor's decoded feedback $\hat{\mathbf{x}}_{k+H} = \mathbf{C}\mathbf{z}_{k+H}^P$ occupy the last n coordinates of the history block of the respective augmented states, so their difference is a coordinate sub-vector of $\hat{\boldsymbol{\xi}}_{k+H} - \boldsymbol{\xi}_{k+H}^\star$. Since a coordinate sub-vector has at most the norm of the full vector, Theorem 1 gives directly, with no additional assumption,

$$\|\hat{\mathbf{x}}_{k+H} - \mathbf{x}_{k+H}\|_2 \le L_{\mathrm{F}}^H \|e_k\|_2 + \sum_{i=0}^{H-1} L_{\mathrm{F}}^{H-1-i}\, \delta_{k+i}^\star. \tag{43}$$

Remark 3 (Scope of the analysis)**.** *The analysis is finite-horizon and concerns the consistency of the learned predictor and the propagation of its prediction error; the local linearization accuracy of the RTI step is treated separately in Section VI. It does not establish global closed-loop stability, a common Lyapunov function, terminal invariant sets, or recursive feasibility under learned uncertainty. The regularity constants used for the empirical assessment are operating-region diagnostics rather than global certificates, all bounds concern the rolled-out latent predictor rather than true-plant tracking error, and $\mathbf{A}(\mathbf{c})$ is a latent linearization parameterization rather than an exact Koopman invariant-subspace operator.*

VI. Real-Time Predictive Control

In this section, we embed the learned history-conditioned predictor into a receding-horizon controller. The prediction rollout propagates the lifted state while updating the scheduling context from decoded state histories, and the optimization is approximated by one real-time-iteration (RTI) step per control update [43], [44] to reduce the online computation of each update; whether a given sampling constraint is met is an empirical question.

A. History-Consistent Scheduled Prediction

At control instant k the controller is initialized from the measured history and predicts over a horizon N according to

$$\begin{aligned}
\hat{\mathbf{X}}_{k|k} &= \mathbf{X}_k, \qquad \mathbf{U}_{k|k}^{\mathrm{in}} = \mathbf{U}_k^{\mathrm{in}}, \\
\mathbf{z}_{k|k} &= g_\theta(\mathbf{X}_k), \\
\mathbf{c}_{k+i|k} &= C_\phi\big(\hat{\mathbf{X}}_{k+i|k}, \mathbf{U}_{k+i|k}^{\mathrm{in}}\big), \\
\mathbf{z}_{k+i+1|k} &= \mathbf{A}(\mathbf{c}_{k+i|k})\, \mathbf{z}_{k+i|k} + \mathbf{B}\, \mathbf{u}_{k+i|k}, \\
\hat{\mathbf{x}}_{k+i+1|k} &= \mathbf{C}\, \mathbf{z}_{k+i+1|k}, \\
\hat{\mathbf{X}}_{k+i+1|k} &= \mathrm{shift}\big(\hat{\mathbf{X}}_{k+i|k}, \hat{\mathbf{x}}_{k+i+1|k}\big), \\
\mathbf{U}_{k+i+1|k}^{\mathrm{in}} &= \mathrm{shift}\big(\mathbf{U}_{k+i|k}^{\mathrm{in}}, \mathbf{u}_{k+i|k}\big),
\end{aligned} \tag{44}$$

where $i = 0, \dots, N-1$ and $i|k$ denotes prediction step i from control instant k. This is the same *context-re-scheduled, latent-propagated* recursion as the training rollout (24): the lifting encoder is evaluated once per control instant, at the start of the horizon, and the context is re-scheduled at every step from the decoded history rather than re-encoded. At the next sampling instant the controller is re-initialized from the newly measured history, which does not re-run the encoder inside the horizon. An alternative is to re-encode the predicted history at each prediction stage (RTI-RE), which introduces additional Transformer encoder evaluations within the horizon; the proposed propagation strategy avoids these repeated encoder evaluations.

B. One-Iteration RTI

Writing $\mathbf{U}_{\mathrm{s}} = \mathrm{col}(\mathbf{u}_{k|k}, \dots, \mathbf{u}_{k+N-1|k}) \in \mathbb{R}^{Nm}$ for the stacked input sequence over the horizon, we solve the receding-horizon problem

$$\begin{aligned}
\min_{\mathbf{U}_{\mathrm{s}}} \quad & \sum_{i=1}^{N} \big\|\hat{\mathbf{x}}_{k+i|k} - \mathbf{x}_{k+i}^{\mathrm{ref}}\big\|_{\mathbf{Q}}^2 \\
& + \sum_{i=0}^{N-1} \Big(\big\|\mathbf{u}_{k+i|k}\big\|_{\mathbf{R}}^2 + \big\|\mathbf{u}_{k+i|k} - \mathbf{u}_{k+i-1|k}\big\|_{\mathbf{S}}^2\Big),
\end{aligned} \tag{45}$$

subject to the componentwise input box $\mathbf{u}_{\min} \le \mathbf{u}_{k+i|k} \le \mathbf{u}_{\max}$. In the input-rate term the predecessor is the previously

applied input, $\mathbf{u}_{k-1|k} = \mathbf{u}_{k-1}^{\mathrm{app}}$, which is a known constant rather than a decision variable, and $\mathbf{Q}, \mathbf{R}, \mathbf{S}$ are positive semidefinite weights. Because the scheduled context depends on the predicted history, the predicted states in (45) depend on the whole input sequence, so this problem is generally nonlinear and is not solved as a convex program; it is handled locally, as follows. No terminal cost, terminal set, state constraint, or robust tube is imposed, and none is claimed below.

One RTI step per sampling instant for (45) consists of: (i) warm-starting the input sequence from the previous solution shifted by one step; (ii) rolling out the nominal trajectory (44) at that sequence; (iii) evaluating, along the nominal trajectory, the sensitivities

$$\begin{aligned} \mathbf{F}_i &= \frac{\partial \mathbf{F}_\Theta}{\partial \boldsymbol{\xi}}\Big|_{(\bar{\boldsymbol{\xi}}_i, \bar{\mathbf{u}}_i)}, \qquad \mathbf{G}_i = \frac{\partial \mathbf{F}_\Theta}{\partial \mathbf{u}}\Big|_{(\bar{\boldsymbol{\xi}}_i, \bar{\mathbf{u}}_i)}, \\ \delta\boldsymbol{\xi}_{i+1} &= \mathbf{F}_i\, \delta\boldsymbol{\xi}_i + \mathbf{G}_i\, \delta\mathbf{u}_i, \end{aligned} \tag{46}$$

and composing them along the rollout into the sensitivity of the predicted outputs to the *whole* input sequence; this retains the dependence of every scheduled context on the inputs applied before it, and does not treat $\mathbf{A}(\mathbf{c}_i)$ as a fixed matrix; (iv) solving one local quadratic subproblem in the increments $\delta\mathbf{U}_{\mathrm{s}}$ around the nominal value $\bar{\mathbf{U}}_{\mathrm{s}}$,

$$\begin{aligned} \min_{\delta\mathbf{U}_{\mathrm{s}}} \quad & \tfrac{1}{2}\, \delta\mathbf{U}_{\mathrm{s}}^\top \mathbf{H}_{\mathrm{RTI}}\, \delta\mathbf{U}_{\mathrm{s}} + \mathbf{g}_{\mathrm{RTI}}^\top \delta\mathbf{U}_{\mathrm{s}}, \\ \text{s.t.} \quad & \mathbf{u}_{\min} - \bar{\mathbf{U}}_{\mathrm{s}} \le \delta\mathbf{U}_{\mathrm{s}} \le \mathbf{u}_{\max} - \bar{\mathbf{U}}_{\mathrm{s}}, \end{aligned} \tag{47}$$

in which the quadratic data $\mathbf{H}_{\mathrm{RTI}}$ and $\mathbf{g}_{\mathrm{RTI}}$ are given by (48); and (v) applying only the first input, $\mathbf{u}_k^{\mathrm{app}} = \bar{\mathbf{u}}_k + \delta\mathbf{u}_k$.

Let $\hat{\mathbf{x}}_{k+1:k+N} \in \mathbb{R}^{Nn}$ denote the stacked predicted states $\hat{\mathbf{x}}_{k+1|k}, \dots, \hat{\mathbf{x}}_{k+N|k}$ obtained by rolling out (44) at the nominal sequence $\bar{\mathbf{U}}_{\mathrm{s}}$, and let $\mathbf{J}_x = \partial\hat{\mathbf{x}}_{k+1:k+N}/\partial\mathbf{U}_{\mathrm{s}}$ denote its sensitivity to the *whole* input sequence, obtained by composing the sensitivities (46) along the rollout, so that $\mathbf{J}_x$ differentiates the scheduled contexts, the low-rank modulation, the latent propagation, and the history shifts, rather than treating $\mathbf{A}(\mathbf{c}_i)$ as a fixed matrix. Writing the input-rate term of (45) as $\mathbf{D}\mathbf{U}_{\mathrm{s}} + \mathbf{b}$, where $\mathbf{D}$ is the first-difference operator and $\mathbf{b} = [-\mathbf{u}_{k-1}^{\mathrm{app}}; \mathbf{0}]$ carries the previously applied input, the quadratic data of (47) are

$$\begin{aligned} \mathbf{H}_{\mathrm{RTI}} &= 2\big(\mathbf{J}_x^\top \mathbf{Q}\mathbf{J}_x + \mathbf{R} + \mathbf{D}^\top\mathbf{S}\mathbf{D}\big), \\ \mathbf{g}_{\mathrm{RTI}} &= 2\big[\mathbf{J}_x^\top \mathbf{Q}(\hat{\mathbf{x}}_{k+1:k+N} - \mathbf{x}_{k+1:k+N}^{\mathrm{ref}}) \\ &\qquad + \mathbf{R}\bar{\mathbf{U}}_{\mathrm{s}} + \mathbf{D}^\top\mathbf{S}(\mathbf{D}\bar{\mathbf{U}}_{\mathrm{s}} + \mathbf{b})\big], \end{aligned} \tag{48}$$

Here $\mathbf{Q}, \mathbf{R}, \mathbf{S}$ are lifted block-diagonally over the horizon as implied by (45); this is the Gauss–Newton form of the linearized scheduled problem, in which the Jacobian is exact for the scheduled rollout while the second-order dependence of the rollout map on the inputs is not retained. The implementation evaluates the objective (47) directly and minimizes it numerically under the increment box constraints with a limited-iteration quasi-Newton method, instead of assembling $\mathbf{H}_{\mathrm{RTI}}$; the iteration budget is fixed a priori. For positive semidefinite weights, $\mathbf{H}_{\mathrm{RTI}}$ is symmetric and positive semidefinite, and it need not be positive definite.

Because the bounds act on the increments, $\delta\mathbf{u} = \mathbf{0}$ is feasible for this subproblem whenever the warm-start nominal input lies inside the box. This is feasibility of the linearized subproblem only: it does not imply recursive feasibility of the nonlinear problem, and no claim is made here of strict convexity, of a unique minimizer, of exact minimization, or of global SQP convergence.

The accuracy of step (iii) is characterized by the local second-order remainder of the augmented map. Treating $\mathbf{F}_\Theta$ as a function of the joint variable $\eta = \mathrm{col}(\boldsymbol{\xi}, \mathbf{u})$, let $\bar{\eta}_i = \mathrm{col}(\bar{\boldsymbol{\xi}}_i, \bar{\mathbf{u}}_i)$ and $\delta\eta_i = \mathrm{col}(\delta\boldsymbol{\xi}_i, \delta\mathbf{u}_i)$ denote the nominal point and the joint perturbation.

Assumption 3 (Local second-order regularity of the augmented map)**.** *The nominal point $\bar{\eta}_i$ and the perturbation segment $\{\bar{\eta}_i + s\,\delta\eta_i : \ s \in [0,1]\}$ lie in a common region on which $\mathbf{F}_\Theta$ is twice continuously differentiable, and on that segment the second derivative is bounded by $\|\mathrm{D}^2\mathbf{F}_\Theta\|_2 \le M_{\mathrm{F}}$, where $\|\cdot\|_2$ denotes the operator norm of the second-derivative bilinear form. The remainder below is used only for perturbations whose segment lies within one such region: it is a local statement, and the same estimate is not asserted for a segment that crosses a boundary on which $\mathbf{F}_\Theta$ is not twice differentiable.*

Under Assumption 3 the expansion

$$\begin{aligned} \mathbf{F}_\Theta(\bar{\eta}_i + \delta\eta_i) &= \mathbf{F}_\Theta(\bar{\eta}_i) + \mathrm{D}\mathbf{F}_\Theta(\bar{\eta}_i)\, \delta\eta_i + \mathcal{R}_i, \\ \mathrm{D}\mathbf{F}_\Theta(\bar{\eta}_i)\, \delta\eta_i &= \mathbf{F}_i\, \delta\boldsymbol{\xi}_i + \mathbf{G}_i\, \delta\mathbf{u}_i, \end{aligned} \tag{49}$$

has the integral remainder

$$\mathcal{R}_i = \int_0^1 (1-s)\, \mathrm{D}^2\mathbf{F}_\Theta\big(\bar{\eta}_i + s\delta\eta_i\big)\big[\delta\eta_i, \delta\eta_i\big]\, \mathrm{d}s, \tag{50}$$

where $\mathrm{D}^2\mathbf{F}_\Theta[\cdot,\cdot]$ is the second-derivative bilinear form, and bounding the norm of the integral by the integral of the norm gives

$$\|\mathcal{R}_i\|_2 \le \frac{M_{\mathrm{F}}}{2}\big\|\delta\eta_i\big\|_2^2. \tag{51}$$

This bounds the model error of the local linearization used in step (iii), and it is local to the region of Assumption 3: it is a property of the linearization itself, and it neither bounds the performance of a single RTI iteration nor establishes SQP convergence, recursive feasibility, or closed-loop stability.

VII. Experiments

This section reports a training-schedule ablation of the history-conditioned predictor, followed by the empirical diagnostics that assess the assumptions and quantities of the consistency analysis of Section V.

A. Training Schedule Ablation

We assess how the training schedule affects the deployment accuracy of the history-conditioned predictor. All variants share the same architecture, data, optimizer, rollout horizon ($H_R = 20$), learning rate, batch size, and scheduled-rollout-stage epoch budget. They differ in which parameters the scheduled-rollout stage optimizes: strategy A optimizes only the low-rank modulation $(\phi, \mathbf{U}, \mathbf{V}, \gamma_\psi)$ while freezing the temporal lifting θ, the nominal transition $\mathbf{A}_0$, and the input matrix $\mathbf{B}$; strategy B freezes only $\mathbf{B}$; strategy C optimizes

TABLE I
TRAINING-SCHEDULE ABLATION: VALIDATION ROLLOUT RMSE UNDER SCHEDULED PROPAGATION, AVERAGED OVER THREE SEEDS ($H_R = 20$ DURING TRAINING). THE STANDARD DEVIATION AT $H = 20$ IS AT MOST 0.0011 FOR B–D.

	Scheduled-rollout stage	$H = 1$	$H = 5$	$H = 10$	$H = 20$
A	modulation only	0.0276	0.1787	0.4300	1.1533
B	all but $\mathbf{B}$	0.0173	0.0271	0.0337	0.0440
C	all parameters	0.0115	0.0228	0.0301	0.0415
D	all, no pretraining	0.0112	0.0217	0.0289	0.0401

all parameters; strategy D additionally omits the nominal one-step pretraining stage and optimizes all parameters from a random initialization. Table I reports the validation rollout RMSE under the deployment-consistent scheduled propagation, averaged over three seeds.

Three observations follow. First, what matters is which parameters the scheduled-rollout objective is allowed to adapt: optimizing only the low-rank modulation while freezing the temporal lifting and the nominal transition (A) degrades the $H = 20$ rollout error by more than an order of magnitude (1.1533 versus 0.0415), so the representation and the nominal transition must adapt together with the modulation. Second, freezing the input matrix brings no benefit: variant B is worse than variant C at every horizon and in every seed, with a 5.9% higher $H = 20$ rollout RMSE and a 24% higher rollout MSE accumulated over the whole 20-step rollout. Third, the nominal pretraining stage is not required: optimizing all parameters directly under the scheduled-rollout objective (D) matches or slightly exceeds C while removing one training stage and reducing the total training time (311 s versus 338 s). We therefore treat the two-stage schedule of Section IV as an optimization initialization rather than a necessity of the model. All remaining results use the deployment predictor described in Section IV, which was trained with the same schedule as variant C; the ablation variants isolate the contribution of each element of that schedule. The separation of B from C and D exceeds the seed-to-seed standard deviation, whereas C and D are statistically indistinguishable.

We also verify that the deployment failure is not an artifact of unbounded latent dynamics. Training with an explicit spectral constraint $\rho(\mathbf{A}(\mathbf{c})) \leq 0.99$ under the one-step objective attains an $H = 20$ rollout RMSE of 1.5681, whereas the same constrained model trained under the scheduled-rollout objective attains 0.0412 (both single-seed comparisons). Enforcing spectral stability therefore does not repair the failure, whereas changing the training objective does.

B. Mechanism and Theory Diagnostics

Table II reports the diagnostics measured on the frozen deployment predictor of Section IV (all parameters optimized under the scheduled-rollout objective, $r = 2$, scheduled propagation) on the soft-robot validation set ($P = 8$, $n = 2$, $m = 2$, $d = 10$). These diagnostics are empirical measurements on the held-out validation distribution: they assess the consistency of the theoretical assumptions and quantities with the operating data, rather than certify global worst-case constants over the entire operating set Ω.[1]

TABLE II
THEORY DIAGNOSTICS ON THE FROZEN SCHEDULED-PROPAGATION PREDICTOR.

Quantity	Role	Measured ($H = 1/5/10/20$)
ϵ_k (absolute)	Prop. 1	median 0.345; q95 0.567
$d^{\mathrm{abs}}(H)$	Prop. 1	0.346/1.122/1.645/2.663
$d_{\mathcal{M}}(H)$	Def. 3	0.120/0.390/0.574/0.928
$\|\mathbf{A}_{k+H-1} \cdots \mathbf{A}_k\|_2$	Prop. 1	1.046/1.247/1.541/2.304
$\|\partial \mathbf{F}_\Theta / \partial \boldsymbol{\xi}\|_2$ ($\widehat{L_{\mathrm{F}}}$)	Assumption 2	median 1.445; max 1.452
Prop. 1 bound / actual	sanity	1.64/2.77/4.22/6.52×
$\widehat{M_{\mathrm{F}}}$ (fitted, δ^2-scaled)	(51)	≈ 0.007
physical RMSE	(43)	0.011/0.022/0.030/0.042
RTI runtime	deployment	p50 78 / p95 120 ms

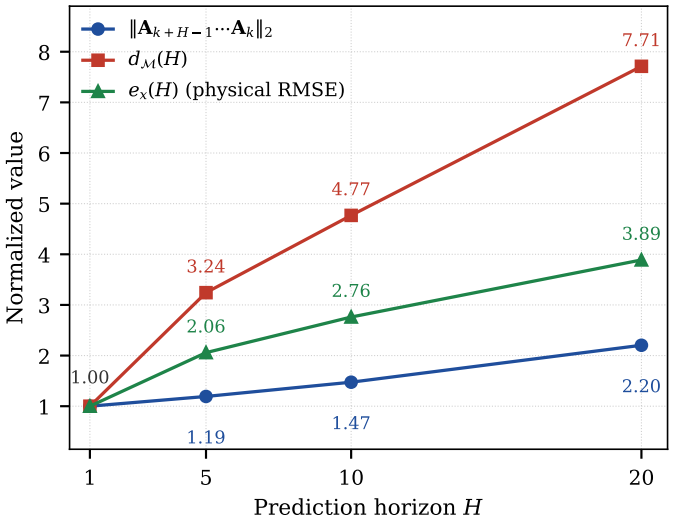


Fig. 3. Normalized growth with the prediction horizon H of the finite transition-product norm $\|\mathbf{A}_{k+H-1} \cdots \mathbf{A}_k\|_2$, the latent rollout defect $d_{\mathcal{M}}(H)$, and the physical prediction error $e_x(H)$; each curve is divided by its value at $H = 1$ so that different units share one axis. The three quantities grow together with H, which is *consistent with* the error-propagation mechanism of Proposition 1. The figure is an empirical illustration and does not by itself establish a causal chain.

Three observations follow, illustrated in Fig. 3. First, the transition-product norm grows from 1.05 to 2.30 (no contraction) in step with $d_{\mathcal{M}}$, numerically supporting Remark 1. Second, the empirical Lipschitz constant is $\widehat{L_{\mathrm{F}}} \approx 1.45 > 1$, so the finite-horizon form of Theorem 1 is the applicable statement and the infinite-horizon limit $\epsilon/(1-L_{\mathrm{F}})$ is not available. Third, the Proposition 1 bound is always $\geq$ the actual defect and only 1.64–6.52× conservative when compared like-for-like (absolute one-step defects ϵ_k against absolute rollout defects d^{abs}), and the fitted $\widehat{M_{\mathrm{F}}} \approx 0.007$ is stable across perturbation scales, supporting (51). The physical-space bound (43) is valid but very loose (using $\widehat{L_{\mathrm{F}}} = 1.445$, 53–47 750× over $H = 1, \ldots, 20$), because $\widehat{L_{\mathrm{F}}}^H$ grows geometrically while the true error saturates; we report this honestly rather than

[1] ϵ_k and d^{abs} are evaluated on all admissible validation windows (stride one); the H-step rollout statistics $d_{\mathcal{M}}$ and the transition-product norms use a stride-four subsample of the same trajectories. All values are medians unless a quantile is stated.

presenting the bound as tight.

These measurements bear on the claim that one-step training insufficiently constrains the rollout distribution. That claim rests on the *conjunction* of four observations—a positive one-step consistency defect ($\epsilon_k > 0$), a rollout defect $d_{\mathcal{M}}(H)$ that grows with the horizon, the failure of naive latent propagation, and the recovery obtained under rollout-aware training—and no single observation is sufficient on its own; the latter two are quantified in the experimental study.

VIII. Conclusion

This paper presented a history-conditioned Transformer–Koopman framework for predictive modeling and control of nonlinear systems whose measurements need not fully characterize the underlying state. The proposed approach combines state-preserving temporal lifting with low-rank modulation of a nominal lifted transition, allowing the predictor to adapt to recent state–input histories while retaining linear latent propagation for a fixed context. To address the mismatch between one-step learning and recursive prediction, a two-stage training procedure was developed, incorporating rollout-aware fine-tuning with context updates along predicted trajectories. A finite-horizon analysis characterized latent rollout inconsistency and prediction-error propagation under explicit assumptions. The learned predictor was further integrated into a one-iteration RTI-MPC formulation that propagates lifted states and updates contexts without repeatedly evaluating the Transformer encoder within the prediction horizon. The resulting framework provides a structured approach to history-dependent prediction and its integration into real-time predictive control, while closed-loop stability and robustness guarantees under model uncertainty remain outside the scope of the present analysis.

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